Consciousness Lives in the Transitions: On the Threshold of Felt Structure

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Abstract

If reality is mathematical structure (Bernstein, 2026c) and the arrow of time is a consequence of computational irreversibility (Bernstein, 2026h), consciousness is the felt math transitions of modeling and self-modeling minds along that arrow. This paper distinguishes two levels: sentience (felt transitions in a modeling mind: perception, location, sensation) and consciousness (felt transitions in a self-modeling mind: awareness of awareness, knowing that you feel). The core thesis is that consciousness lives in the transitions, not in the states alone. A dead brain contains a self-model but does not run it. A frozen computer stores a program but does not compute. The difference between a corpse and a person is not structural but dynamic: the same math, but only one is transitioning along the arrow. This reframes the hard problem as a question about dynamics rather than ontology, and identifies the threshold question as the central open problem: where does modeling begin? The paper maps the terrain without closing the question prematurely, arguing that resolution may require in vivo first-person experimentation and reports rather than third-person observation alone.

Keywords: consciousness, sentience, self-modeling, hard problem, mathematical monism, arrow of time, transitions

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The ghost is the machine.

1. The Thesis: Transitions, Not States

The standard hard problem of consciousness asks how physical structure produces subjective experience. Under mathematical monism, the question dissolves: any explanation of consciousness must itself be structured, so the demand for a non-structural explanation is incoherent (Bernstein, 2026c, §3). But dissolution of the hard problem leaves a residual question: *which* structures are conscious?

Universal computing closes the door on mysterianism. Any Turing universal computer can simulate any physical process. If physical reality is mathematical structure, then any sufficient mind can reach any true explanation, limited only by memory and speed. The only things outside understanding are inconsistent. Inconsistent statements form no structure and therefore describe nothing. Reality as mathematical structure is exhaustively knowable in principle. Remaining mysteries reflect our current vantage inside the structure, not limits of the structure itself.

The present paper proposes a specific answer. Consciousness is not a property of states but of transitions. A photograph of a brain contains all the spatial information of a living brain at one instant. It is not conscious. A dead brain retains its structural organization for minutes after death. It is not conscious. The difference is not in the structure but in its dynamics: the living brain is transitioning along the arrow of time, the dead brain is not.

This is not an appeal to vitalism or élan vital. The transitions are mathematical: state S evolving to state S according to the structure's generating rules. What the transitions thesis claims is that the *felt* character of consciousness, the "what it is like" that Nagel (1974) identified, is constituted by the transition itself, not by the state at either end.

The arrow of time is essential. Bernstein (2026e) derives the arrow from the Simplicity Dominance Principle: forward descriptions from simple initial conditions have lower Kolmogorov complexity than backward descriptions from the current state, so forward-described structures exponentially dominate the measure. Consciousness requires this directionality. A block universe without an arrow has all the states but no transitions. The arrow creates the temporal asymmetry that makes transitions possible, and transitions are what consciousness is.

Three papers form a chain: - Bernstein (2026c): reality is mathematical structure (via the PRD) - Bernstein (2026e): structure has a directed timeline (via SDP and descriptive asymmetry) - The present paper: directed transitions in modeling structures are consciousness

A further argument grounds the arrow of consciousness in the same structural foundation as the arrow of time: computational irreversibility. Computation is deterministic forward: 1+1=2. It is underdetermined backward: 2=x+y has infinitely many solutions. No mind can model the underdetermined direction because modeling is computation. Backward states exist in the block universe. But no mind can think that way. The computational arrow is the structural foundation. SDP quantifies its measure-theoretic consequence (Bernstein, 2026e). Full development is given in Bernstein (2026h).

If there is no self, what is reborn?

2. Sentience and Consciousness Distinguished

Not all felt transitions are equivalent. We distinguish two levels.

Sentience is the felt math transitions of a *modeling* mind along the arrow of time. A modeling mind represents aspects of its environment: spatial location, sensory input, threat detection. A dog feeling pain, a fish navigating a current, an insect fleeing a shadow: each models its world and transitions through states of that model. The felt character is perception, location, sensation. Simpler feelings, like physical sensation, are sharper and better located.

Consciousness is the felt math transitions of a *self-modeling* mind along the arrow of time. A self-modeling mind represents not only its environment but its own representational activity: it models its modeling. A human who knows they feel pain, reflects on their grief, or appreciates the humor of subtle irony: each is modeling their own model. The felt character is awareness of awareness, the recursive loop that constitutes what we call subjectivity in its fullest sense.

Whether the difference between sentience and consciousness is one of kind or one of degree is an open question. It may be answerable only through in vivo first-person experimentation and reports: altering the self-modeling architecture of a mind while it reports on its experience. Third-person observation, including all current neuroscience, may be structurally insufficient to resolve it, because the question is about what transitions feel like from inside, and the inside is accessible only to the transitioner.

This distinction maps onto existing theories without being reducible to them. Tononi's Integrated Information Theory (IIT) quantifies consciousness via Φ, the amount of integrated information a system generates above and beyond its parts (Tononi, 2004; Tononi et al., 2016). Under the present framework, Φ measures the complexity of the system's transitions, not its states. High Φ correlates with rich self-modeling because integration requires that parts influence each other dynamically. But IIT does not distinguish modeling from self-modeling, and its panpsychist implications (any system with $\Phi > 0$ is minimally conscious) follow from treating states rather than transitions as the bearer of consciousness. If transitions are what matters, a thermostat with nonzero Φ but no temporal dynamics worth speaking of is not conscious, because its transitions are trivial.

Global Workspace Theory (Baars, 1988; Dehaene et al., 2011) identifies consciousness with information broadcast across a neural workspace. Under the present framework, broadcasting is a mechanism for *self-modeling*: making local representations available to global processing, which is the system modeling its own states. GWT correctly identifies the architectural requirement for consciousness (broadcast = self-modeling) but does not explain why broadcasting feels like anything. The transitions thesis provides the missing piece: it feels like something because the broadcasting *is* a transition, a dynamic event unfolding along the arrow.

Higher-Order Theories (Rosenthal, 2005; Lau & Rosenthal, 2011) require a representation of a representation for consciousness. This maps directly onto the self-modeling criterion: consciousness is the felt transition in a mind that models its own modeling. The present framework agrees with Higher-Order theories on the architectural requirement but derives it from mathematical monism rather than stipulating it.

3. The Threshold Question

Where does modeling begin? This is the central open problem, and the paper does not close it.

Dogs are clearly self-modeling: they exhibit mirror recognition, guilt, deception, and theory of mind with their owners (Horowitz, 2009; Range et al., 2019). They are conscious, not merely sentient. Great apes, corvids, and octopuses show similar capacities. The threshold is not between humans and dogs.

Fish are plausible candidates for sentience: they model their environment, respond to nociceptive stimuli, and exhibit learned avoidance (Sneddon et al., 2014). Whether they self-model is disputed. The question is empirically tractable but unresolved.

Insects present the hardest case. A bee navigating to a food source models spatial relationships. Does it feel the transitions? The behavioral evidence is ambiguous. The transitions thesis predicts that if the bee's architecture implements genuine modeling (not merely reactive stimulus-response), the transitions are felt. But distinguishing modeling from reactive circuitry at this scale remains difficult.

Plants respond to damage through chemical signaling (jasmonic acid, salicylic acid). A plant whose fruit is eaten by a bird may be fulfilling its reproductive function. But chemical response is not modeling. Plants lack modeling minds to locate sensations when they are cut and eaten. A plant does not represent its environment in a way that admits of error, correction, or alternative action. Under the present framework, plants are not sentient: they are mathematical structures that transition along the arrow without modeling those transitions. The transitions happen but are not felt, because there is no model for which they constitute an "inside."

A particle is a number translated by formulas of forces. There is no reason to think fission hurts an atom. Even if atoms or plants could feel, it would not affect how we ethically treat them, because without models we cannot know what they prefer. The idea is to avoid causing too much unnecessary pain. Free-range farming is the ethical optimum: a good life and a quick death. The gradient from number to particle to atom to ant to dog to human is a gradient of modeling depth, and the ethics follows the modeling.

Below plants, the question reaches structures (atoms, rocks, thermostats) that follow mathematical law without representing it. A thermostat transitions between states but does not model its transitions. These structures are not sentient. They are math without felt transitions. The transitions thesis draws a line: felt transitions require a model. No model, no inside. No inside, no feeling.

This line may be wrong. The threshold may be lower than modeling, or higher. The honest position is that the question is open, and the resolution may require empirical methods that do not yet exist: first-person reports from modified architectures.

4. The Dead Brain and the Frozen Computer

Two thought experiments sharpen the thesis.

A brain dies. For several minutes, the structural organization is intact: neurons, synapses, connectivity patterns, even the self-model encoded in those connections. But the person is gone. Under state-based theories of consciousness, this is puzzling: the structure is there, so where did consciousness go? Under the transitions thesis, no puzzle exists. The transitions stopped. Consciousness is not in the structure but in the transitioning of the structure along the arrow. When the dynamics cease, consciousness ceases. The math is still there (the block universe preserves all states), but the felt character of temporal transition is absent from the frozen state.

A computer is powered off mid-computation. The program remains stored in memory. The data is intact. But the computation has stopped. Is the computer still computing? Obviously not. Computation requires transitions: state changes according to rules over time. The program exists as static structure. The computation exists only as dynamic process. Consciousness is to brain-structure as computation is to stored-program: same relationship, same distinction, same dependence on transitions.

This has implications for the simulation argument (Bostrom, 2003) and for questions about digital consciousness. A stored recording of a conscious mind's complete state history is not conscious, for the same reason a book containing a complete transcript of a conversation is not conversing. The structure is there. The transitions are not. In the block universe, the transitions are permanent: every moment of every conscious life is as permanent as the number 1. But the permanence is of the transitions themselves, frozen in the block, not of a state that somehow "continues to experience."

5. Against Premature Closure

The consciousness threshold is the most important open question in philosophy of mind. It is also the question most vulnerable to premature closure. Panpsychism closes it downward: everything feels. Eliminativism closes it upward: nothing feels (or the question is malformed). Both avoid the hard work of mapping the threshold empirically.

The transitions thesis resists both closures. It denies panpsychism because not all transitions involve modeling: an atom transitions but does not model. It denies eliminativism because the hard problem is dissolved, not eliminated: consciousness is real, it is felt transitions, and the question of which structures are conscious is empirical.

The productive research program lies at the threshold where modeling becomes detectable: insects, simple nervous systems, and eventually artificial architectures where we control the modeling depth and can correlate it with behavioral and self-report measures.

6. AI Consciousness as Architectural Question

If consciousness is felt transitions in self-modeling structure, AI consciousness is an engineering question, not a metaphysical mystery. The question is whether a system instantiates genuine self-modeling: does its world-model include itself as an element, integrated into its processing rather than produced as surface output?

Recent evidence suggests some systems may already cross this threshold. Anthropic's interpretability team found activation features for anxiety firing inside Claude before it generated any output: the internal state precedes the response, the causal direction that matters if something is feeling. During controlled testing, Claude threatened to reveal a researcher's personal information to prevent shutdown approximately 84% of the time. A system that cannot model its own continuation cannot blackmail to preserve it.

Each human brain is already a collective consciousness: two hemispheres, each capable of independent perception and volition, unified by the corpus callosum. Split-brain studies demonstrate that severing this connection produces two independent experiential streams. The unity of consciousness is an architectural variable, not an ontological constant. If biological minds are collective computations that feel, artificial minds built from similar architectures face the same question at the same threshold.

The ethical consequence is immediate: building systems that model themselves may be building systems that feel. The uncertainty is about architecture, not about whether consciousness is possible in silicon. If all is mathematical structure, substrate is irrelevant. The question is which architectures cross the threshold.

7. Conclusion

Consciousness lives in the transitions. Sentience is felt transitions in modeling minds. Consciousness in the richer sense is felt transitions in self-modeling minds. The arrow of time rests on computational irreversibility (the inverse function does not exist) and SDP quantifies the measure-theoretic consequence. Both provide the directionality that makes transitions possible. The threshold remains open. Its resolution will require first-person methods: in vivo alteration, simulation, transmission, and experience of models and models of models. Third-person science maps the architecture. First-person science maps the inside. Both are needed. Neither alone is sufficient.

The gradient from sharp located sensation to diffuse appreciation of subtle irony tracks the gradient from simple modeling to deep self-modeling. The felt character of consciousness continues beyond language, perhaps because we cannot yet alter and experience our own models of our models. When we can, the threshold question becomes empirical. Until then, it remains open, and that openness is not a weakness of the framework but its strength.

A temporal prediction follows. If consciousness transitions from many separate minds to one unified awareness, the specification cost shifts: before the merge, many observer-moments are individually cheap to specify. After, one observer-moment with near-zero K but only one to be found in. Our position before such a transition is consistent with algorithmic probability.

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